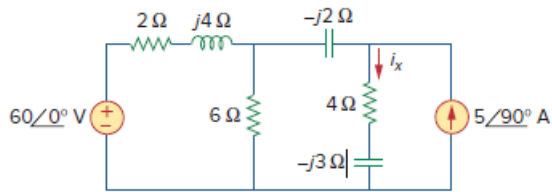


Problem

Use the method of source transformation to find I_x in the circuit of Fig. 10.96.

Figure 10.96:



Step-by-step solution

Step 1 of 6

We transform the voltage source to a current source,

$$I_s = \frac{60 \angle 0^\circ}{2 + j4}$$

$$I_s = 6 - j12$$

The new circuit is shown in Fig-(a),

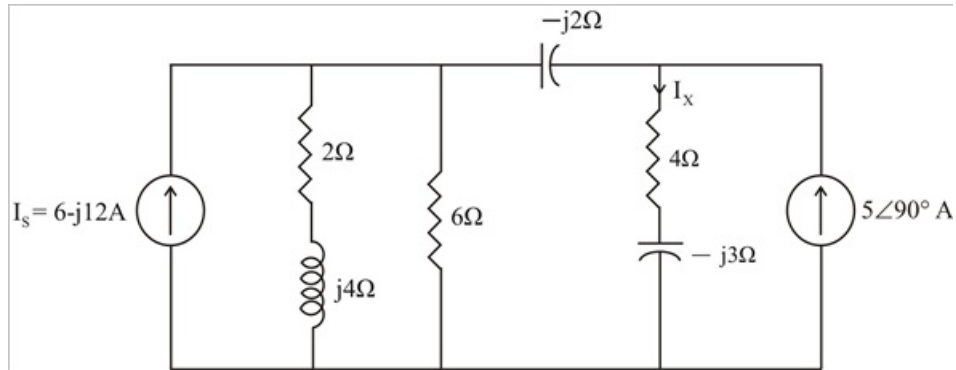


Fig-(a)

Step 2 of 6

From Fig-(a),

$$Z_s = 6 \parallel (2 + j4)$$

$$Z_s = \frac{6(2 + j4)}{8 + j4}$$

$$Z_s = 2.4 + j1.8$$

Therefore,

$$V_s = I_s \times Z_s$$

$$V_s = (6 - j12)(2.4 + j1.8)$$

$$V_s = 36 - j18$$

$$V_s = 18(2 - j)$$

With these we transform the current source on the left hand side of the circuit to A voltage source. We obtained the circuit in Fig-(b).

Step 3 of 6

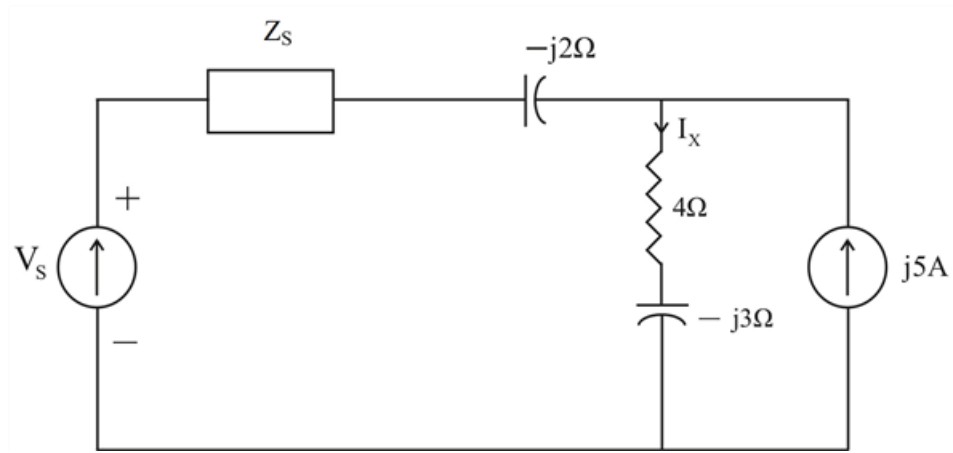


Fig-(b)

Step 4 of 6

From Fig-(b),

$$Z_0 = Z_s - j2$$

$$Z_0 = 2.4 - j0.2$$

$$Z_0 = 0.2(12 - j)$$

Therefore,

$$I_0 = \frac{V_s}{Z_0}$$

$$I_0 = \frac{18(2 - j)}{0.2(12 - j)}$$

$$I_0 = 15.517 - j6.207 \text{ A}$$

Step 5 of 6

With these, we transform the voltage source in Fig-(b) to a current source. We obtain the circuit in Fig-(c)

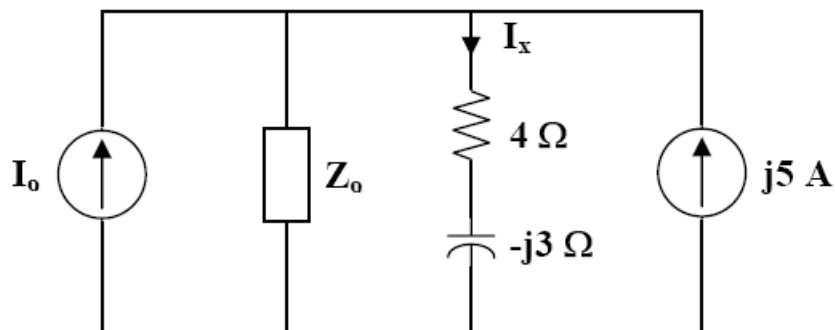


Fig-(c)

Step 6 of 6

From Fig-(c),

By using current division rule we get,

$$I_x = \left(\frac{Z_0}{Z_0 + 4 - j3} \right) \times (I_0 + j5)$$

$$I_x = \left(\frac{2.4 - j0.2}{2.4 - j0.2 + 4 - j3} \right) \times (15.517 - j6.207 + j5)$$

$$I_x = \left(\frac{2.4 - j0.2}{6.4 - j3.2} \right) \times (15.517 - j1.207)$$

$$I_x = (0.3125 + j0.125) \times (15.517 - j1.207)$$

$$I_x = 5 + j1.562$$

$$\therefore \boxed{I_x = 5.238 \angle 17.345^\circ \text{ A}}$$